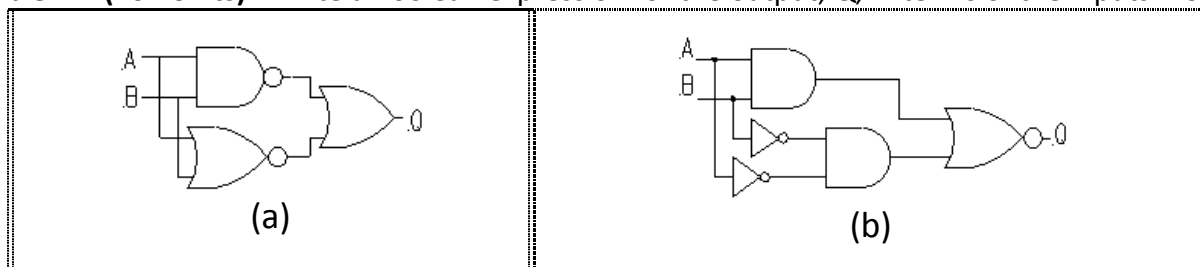


اسم الطالب: _____ موعِد المختبر: _____
اسماء المدرسين: _____
(1) سليمان بني أحمد، (2) باسم الرفاعي، (3) وفاء ضبابات، (4) اصلاح الغرايبة، (5) اسماء الختوم (6) رائد خليل

السؤال الاول (4 علامات): ضع اشارة صح حول الى جانب رقم شعبتك.

اسم المدرس	اليوم	موعد المختبر	رقم الشعبة
اصلاح الغرايبة	ح	10:00 - 12:00	1
رائد محمد خليل	ح	12:00 - 02:00	2
سليمان احمد بني احمد	ن	10:00 - 12:00	3
اصلاح الغرايبة	ن	02:00 - 04:00	4
اسماء محمد الختوم	ث	02:00 - 04:00	5
وفاء ضبابات	ر	08:00 - 10:00	6
وفاء ضبابات	خ	08:00 - 10:00	7
اصلاح الغرايبة	خ	12:00 - 02:00	8
باسم محمد الرفاعي	خ	02:00 - 04:00	9

Problem 1 (10 Points): Write a Boolean expression for the output, Q, in terms of the inputs A & B.

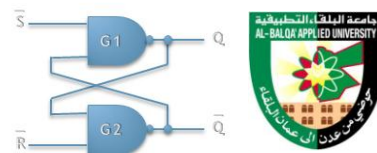


Part a: $Q = f(A, B) =$

Part b: $Q = f(A, B) =$

Problem 2 (20 Points): Simplify the following expressions

$$F = x + x'y$$



$$F = xyz + x'y + xyz'$$

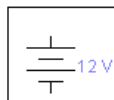
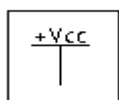
Problem 3 (22 Points): Choose the best answer

1- The function of this icon is :



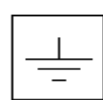
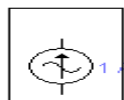
- a) Turn on EWB
- b) Switch OFF/ON the circuit
- c) Give logic 1 to the input
- d) Give logic 0 to the input

2- To connect a circuit with logic 1 we use:



a)

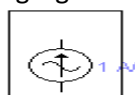
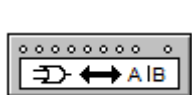
b)



c)

d)

3- To view the truth table of a logic gate we use



a)

b)



c)

d)

4- This icon represent :

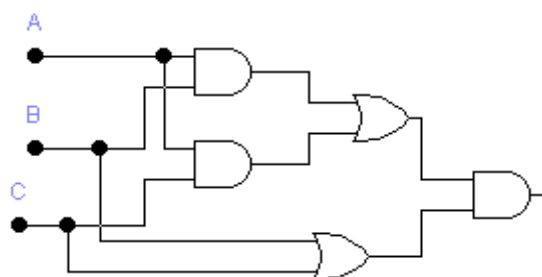


- a) power supply
- b) an input node
- c) an output led
- d) none of the above

5- The logic circuit that all the inputs must be 1 to get 1 on the output is :

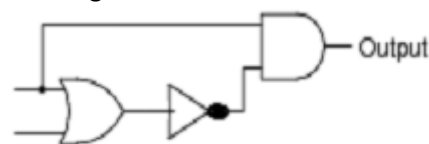
- a) OR
- b) NOR
- c) AND
- d) NAND

6- The Boolean expression of this circuit is:



- a) $(A.B) + (A.B) . (C+B)$
- b) $(A.B) + (A.C) . (B+C)$
- c) $(A+B) . (A+C) + (C.B)$
- d) $(A.B) + (A.C) + (B.C)$

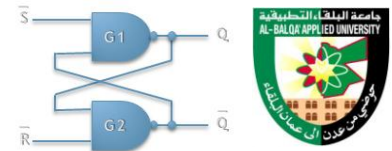
7. The simplified Boolean expression for the next diagram is:



- A. $F = (X' + Y') . X$
- B. $F = 1$
- C. $F = X' . Y' . X$
- D. $F = 0$

8. A four-variable function $f(x,y,z,w)$ has:

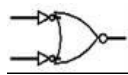
- A. 8 maxterms
- B. 16 maxterms
- C. 8 minterms
- D. 4 maxterms



9. The simplified function that represents $F(A, B, C) = \sum(0, 1, 4, 5)$ is:

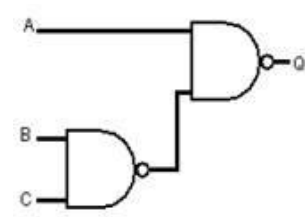
- A. $F = (A' \cdot B') + (A \cdot B')$
- B. $F = B'$
- C. $F = (B' \cdot C') + (B' \cdot C)$
- D. $F = A'$

10. The circuit which is equal to the following circuit is:



- A. OR
- B. AND
- C. XOR
- D. XNOR

11. The correct Boolean expression that represents the next circuit are:

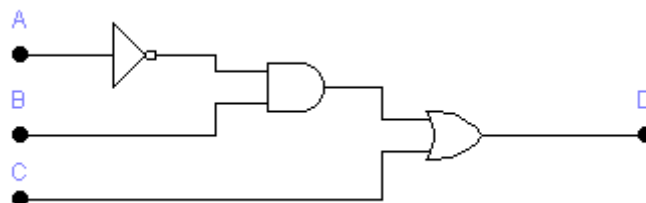


- A. $Q = (A(BC))'$
- B. $Q = A' \cdot B \cdot C$
- C. $Q = A' + B + C$
- D. Both A and C are correct

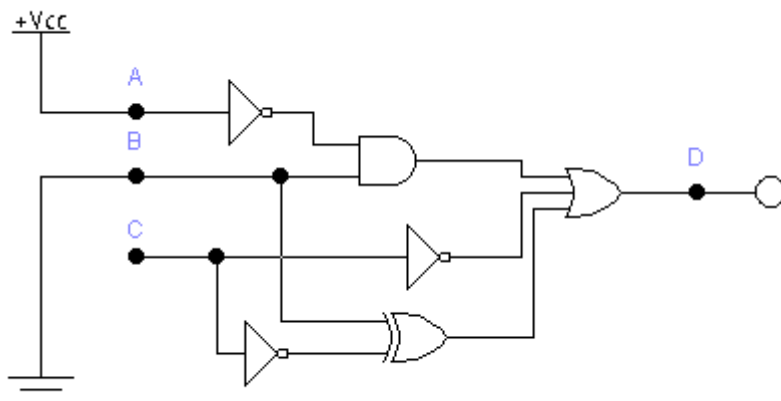
12. The simplification of $F = (x+y)' \cdot y'$ using Demorgan theory is:

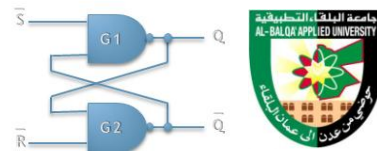
- A. $F = Y'$
- B. $F = X'$
- C. $F = X' Y' + Y'$
- D. $F = X' \cdot Y'$

Problem 4 (5 Points): In the following circuit, if we put $A=1$, $B=1$, and $C=0$ then the output D will be logical zero. What are the combinations of inputs required to have the output (D) to be logical 1.



Problem 5 (5 Points): In the following circuit, the inputs A and B are connected to the V_{CC} source and to the ground, respectively. Where should the third input C be connected so that the probe connected to the output be in the ON state?





Problem 5 (20 Points): Design logic circuit with three inputs (A,B,C) and the output 0 if the binary input is even and 1 if the binary input is odd.